

JOHN PAUL NEWMAN

Lead DevOps Engineer
Senior Software Engineer



johnpaulnewman.com



johnpaul.newman

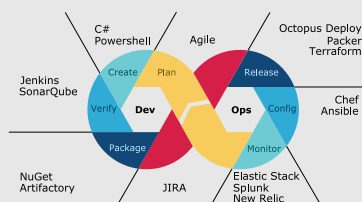


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jpnewman

Skills



Programming Languages

Bash • Python • Golang
C# • SQL • Entity Framework
C • C++ • Powershell

Infrastructure as Code (IaC)

Terraform • Ansible • Chef
Powershell DSC • Packer

Operating Systems

Windows • Linux (Ubuntu)
macOS

Platforms

AWS (EKS, EC2, S3, RDS, MSK, Lambda, Kinesis)
Azure (Service Fabric, App Services, API Management, Service Bus, VMSS, Key Vault, Storage Account)
Docker • Kubernetes • VMware
Xbox 360 • PS3 • iOS • Android

CI/CD

GitHub • GitLab • Argo CD
Azure DevOps • Octopus Deploy
Jenkins • GoCD • Artifactory • Gerrit

Monitoring

Datadog • Splunk • New Relic
AWS CloudWatch
Azure Application Insights
Elastic Stack
Influxdb • Prometheus • Grafana

Alerting

Azure Monitor Alerts
AWS CloudWatch Alarms - SNS
Opsgenie • PagerDuty • Zabbix

Summary

Senior/Lead Software Engineer with Windows and Linux DevOps experience working in an Agile environment for FinTech ISO 9001 and PCI-DSS compliant businesses. Experienced in infrastructure automation, using Terraform, Packer, Ansible, and Chef. Experienced with Amazon Web Services (AWS), Azure, and VMware. Implemented High Availability (HA) and Disaster Recovery (DR) best practices. Designed and managed Continuous Integration (CI)/Continuous Deployment (CD) tools, processes, and pipelines.

Work Experience

Lead DevOps Engineer Divido

Jun 2022 - Present
UK

FinTech - Software as a Service (SaaS) Buy Now, Pay Later (BNPL) platform, providing lenders and merchants connectivity.

Responsibilities: -

- Manage a team of two Senior Site Reliability Engineers (SRE) and one Senior Data Engineer
- Support the AWS infrastructure and CI/CD pipelines in a DevOps capacity
- Support the Platform in SRE/on-call capacity
- Support, patch, and maintain Divido AWS infrastructure, consisting of Amazon Elastic Kubernetes Service (EKS), Amazon Relational Database Service (RDS), Amazon Managed Streaming for Apache Kafka (MSK)
- Modernize Divido's tech stack and infrastructure
- Support ISO9001 efforts and Due Diligence

Key Achievements: -

- Managed a team of Senior SRE and Data Engineers
- Refactored HashiCorp Terraform to reduce change blast radius and allow development of multiple aspects for the infrastructure simultaneously
- Implemented High Availability (HA) and Disaster Recovery (DR) best practices
- Created Terraform HTML Plan and Access Reports
- Migrated from Continuous Delivery (CD) process from GitLab to GitHub Actions
- Developed Bash scripts to manage and diagnose issues with Debezium Change Data Capture (CDC) pipelines
- Designed and implemented Atomic Repositories, where microservices contain all items needed to be independently deployed
- Designed a Hub and Spoke network topology to simplify VPNs and Access Controls
- Developed Platform support tools using GoLang
- Created multiple Jupyter Notebooks to graph and analyze incidents
- Created NodeJS AWS Lambda HTML to PDF rendering function using Chromium, Puppeteer, and ExifTool
- Created Training Videos for developers, 3rd parties and on-boarding new members
- Managed the migration from Chartmuseum to Harbor
- Managed the move from DataDog metrics to Prometheus and Grafana
- Implemented the use for tools like Telepresence for debug and Istio Service Mash plugin Kiali for visualizing internal service communication
- Implemented SFTP, with PGP decryption, using AWS Transfer Family
- AWS Cost Savings

Senior DevOps Engineer RateSetter (acquired by Metro Bank)

Jul 2019 - Jun 2022
UK

FinTech - Award winning provider of peer-to-peer personal loans.

Responsibilities: -

- Automate the deployment, maintenance, and patching of platform infrastructure for all environments
- Implement and support platform logging infrastructure such as Splunk, Azure Application Insights and alerts
- Manage project requirements across multiple stakeholders, including IT Ops, InfoSec, Developers, and Business

Key Achievements: -

- Designed and implemented key 3rd party DocuSign, AccountScore, and Bureau Glass.co.uk infrastructure
- Introduced Terraform for Azure infrastructure with JSON workspace variables
- Created a process for development teams to request infrastructure via Pull Requests (PRs)
- Introduced Terraform infrastructure change sign-off and promotion processes using BitBucket PRs and branches
- Designed and implemented Azure API Management and Service Bus for a key Metro Bank loan origination platform
- Redesigned Azure Service Fabric with Azure best practices
- Implemented Azure App Services for both Website and Azure Functions
- Developed a C# ASP.NET Core website to display historical motor loans backed by an AWS RDS MySQL Database and S3 Buckets
- Created a NuGet library for parsing fixed width Call Dialler Reports (CDRs), including a Cobol PIC (Picture) Clause

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Projects

Motor Loan Website

- C# .NET Core
 - AWS SDK for .NET, SignalR
 - Dapper, Polly
 - Serilog, Application Insights
 - Moq, xUnit
- Azure
 - Azure Active Directory (AAD) Authentication
 - App Service (Web App)
 - App Configuration, Key Vault
 - Event Grid
 - Log Analytics
- AWS
 - AWS RDS MySQL Database
 - S3 Buckets
 - SSH Tunnel from Azure to AWS
- JQuery with AJAX

Data Pipeline

- AWS Lambda (C# & Python)
- AWS Kinesis
- AWS Glue & Athena
- AWS EMR (Scala & PySpark)
- Apache Avro

OCR Solution - Hackathon : Winner

- Python (Flask, Celery, RethinkDB)
- RabbitMQ

Reporting Service

- C# (Entity Framework, AutoMapper)
- JQuery & d3.js

Artifactory S3 Export

- GoLang, AWS S3

URLInfo (Interview Application)

- GoLang

Windows Slipstream

- Powershell, Packer, VirtualBox

Gerrit Backup Solution

- Python, AWS S3

C# Dependencies Graph

- Ruby & JQuery, Graphviz & SVG

Chef LWRP

- Chef & Powershell

Palm OS RSRC Extractor

- C++, MFC

Adobe Photoshop Automation

- C++, COM

Translation Memory Validation

- VBScript, Classic ASP, XSLT with Javascript functions

Qualifications

Certificate in C++ Programming

Open University (MT262 Module)

Certificate in Programming

Cambridge Information Technology
(Pascal & Delphi)

BTEC 1st Diploma in Science & Health

12 Distinctions and 4 Merits

Work Experience

Senior Software Engineer/DevOps

Wonga

Oct 2012 - Jun 2019

Ireland - UK

FinTech - Payday Loan provider.

Responsibilities: -

- Supported the migration of a monolithic application to Service Oriented Architecture (SOA), using NServiceBus MSMQ plus DTC and RabbitMQ transportation layers
- Create AWS infrastructure, including EC2 Windows instances with Active Directory servers, for migration from data center
- Implement and maintain of all CI/CD systems, including Jenkins, Artifactory, Gerrit, Octopus Deploy, and Windows build/deploy agents
- Create Chef and Ansible playbooks for CI/CD servers
- Create HashiCorp Terraform templates to deploy AWS and VMware architecture
- Create HashiCorp Packer templates for AWS AMIs and VMware templates
- Interim Scrum Master and Team Leader
- Supported KPMG Administration in the wind-down of the businesses

Key Achievements: -

- Introduced the use of NuGet, Artifactory, and Octopus Deploy to manage dependencies and package deployments
- Created a process to deploy a legacy VB6 application using Chef, Powershell, and AWS S3
- Created a Data pipeline in Python using AWS Lambda, Kinesis, and Glue.
- Created Ansible roles to deploy an ELK/Elastic stack, Artifactory, Gerrit, and Jenkins servers
- Developed a RESTful Reporting Service using C#, Entity Framework (EF), JQuery and d3.js
- Created Powershell build and deploy scripts to replace, Ruby Rake and MSBuild scripts, which included a DSL implemented in PowerShell for NuGet packing
- Developed a Ruby application to create Graphviz DOT files of Microsoft Visual Studio solution dependencies and display them via a JQuery website that allowed selection and filtering
- Developed and maintained multiple in-house tools using Ruby on Rails, C#, AWS Amazon Simple Queue Service (SQS) and Amazon Simple Notification Service (SNS)

Software Internationalization Engineer II

PopCap Games International (acquired by Electronic Arts)

Aug 2007 - Oct 2012

Ireland

Game Development - Creator of flagship games, like PopCap's Bejeweled, Plants vs. Zombies, Peggle, and Zuma.

Responsibilities: -

- Internationalization (I18N) and localization (L10N) of all flagship games, across multiple platforms (PC, iOS, Xbox 360, PlayStation, Nintendo DS) and in multiple development languages (C++, Objective-C, Flash ActionScript)
- Liaised with US and Asia-Pacific (APAC) project teams to achieve timely delivery of project goals
- Acted as point of contact and Internationalization subject matter expert

Key Achievements: -

- Reduced art resource requirements by automating Adobe Photoshop to create Asian language Bitmap Fonts
- Created a script to import/export text from Photoshop files, allowing for faster translation of graphical assets
- Developed numerous Python, JavaScript, and Visual Basic for Applications (VBA) tools to reduce engineering time and resources
- Managed multiple key projects simultaneously - liaising with multiple project stakeholders in the US and APAC
- Consistently took the initiative to scope new technology requirements and analyse project impacts

Software Internationalization and Build Engineer

Connect Global Solutions (acquired by Welocalize)

2002 - 2007

Ireland

Worked as Lead Engineer on various projects, including for Apple, AOL, and Canon. Responsible for a small team.

Software Localization Engineer

Bowne Global Solutions (acquired by Lionbridge)

2000 - 2002

Ireland

Worked as a Localization Engineer for multiple clients, including Handspring VisorPhone

IT Network Administrator/Software Localization Engineer

Tek Translation

1998 - 2000

UK - Ireland

Migrated Novell to Windows NT 4 and worked as a Localization Engineer for multiple clients

Laboratory Technician/IT Network Administrator

Aylwin Girl's School

1997 - 1996

UK

Maintenance of 250 workstations and Novell servers